

Thesis for obtaining the academic degree
Bachelor of Science

**Probing Fast Radio Burst Environments:
Cross-Correlating Deep and Shallow
FRB Surveys with KiDS DR5**

Will Ruppert
born in Herdecke

2026

Research Group Rhode / Elsässer
Faculty of Physics
TU Dortmund University

Primary reviewer:	PD Dr. Dominik Elsässer
Secondary reviewer:	Prof. Dr. Hendrik Hildebrandt
Submission date:	15th October 2026

Abstract

The abstract is a short summary of the thesis in English. Together with the German summary, both summaries have to fit on this page.

Kurzfassung

Hier steht eine Kurzfassung der Arbeit in deutscher Sprache inklusive der Zusammenfassung der Ergebnisse. Zusammen mit der englischen Zusammenfassung muss sie auf diese Seite passen.

Contents

1	Introduction	1
2	Theoretical Background	2
2.1	Cosmological Background	2
2.2	Friedmann Equations and Cosmic Expansion	4
2.3	Fast Radio Bursts	5
2.4	Galaxy Surveys and Catalogs	9
2.5	Structure Formation	9
2.6	Correlation Functions and Angular Power Spectrum	9
3	Methodology	13
3.1	Redshift Distribution	13
3.2	Background Correlation	13
3.3	Auto-Correlation Analysis	13
3.4	Cross-Correlation Analysis	13
4	Results	14
4.1	Fisher Matrix	14
5	Conclusion & Outlook	15
6	Structure of the thesis	16
6.1	Erstellen des Ausgabedokuments mit Make	17
6.2	Erstellen des Ausgabedokuments mit Texmaker	17
6.3	Zahlen und Einheiten	19
6.4	Das Literaturverzeichnis	20
6.5	Figures	20
6.6	Tables	21
A	Appendix	22
	Bibliography	23

1 Introduction

- relevance
- emerging probes $\rightarrow$ different papers cite
- whats plan: frb, galaxy survey $\rightarrow$ combination of both

2 Theoretical Background

2.1 Cosmological Background

The foundation of modern cosmology is based on the *Cosmological Principle*, which states that the Universe is homogeneous and isotropic on large scales. Homogeneity implies that the large-scale properties of the Universe are statistically uniform across different locations, whereas isotropy means that no direction is preferred. Although this principle is introduced as a fundamental assumption in modern cosmology, it also represents a hypothesis that can be tested through observations. Current astronomical evidence strongly supports the validity of the cosmological principle on the largest observable scales [30, 14].

Based on the cosmological principle, the Universe can be described by the *Friedmann-Lemaître-Robertson-Walker* (FLRW) metric

$$ds^2 = -c^2 dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2 \theta d\phi^2) \right]$$

with the spatial comoving coordinates (r, θ, ϕ) , the speed of light c , and the cosmic time t [3]. The parameter k describes the curvature of the Universe, which can take values of -1 , 0 , or $+1$ for open, flat, and closed geometries, respectively. The scale factor $a(t)$ describes the relative expansion of the Universe and is conventionally normalized to its present value,

$$a(t_0) = 1.$$

For earlier cosmic times, the scale factor satisfies $a(t) < 1$, whereas in an expanding Universe it increases with time and can exceed unity in the future. A static Universe would correspond to $\dot{a}(t) = 0$, resulting in a vanishing Hubble parameter [10].

The expansion rate of the Universe is quantified by the Hubble parameter,

$$H(t) = \frac{\dot{a}(t)}{a(t)},$$

which describes the relative change of the scale factor with cosmic time. At the present time

$$H_0 = H(t_0)$$

is referred to as the Hubble constant [3].

The expansion of space introduces a distinction between physical distances and comoving distances. While the physical distance between two objects changes with time due to the expansion of the Universe, the comoving distance remains constant for objects that follow the general expansion of space. The physical or so called proper distance is related to the comoving distance χ by

$$D(t) = a(t)\chi.$$

Thus, comoving coordinates remove the effect of the cosmic expansion and provide a convenient description of the large-scale structure of the Universe [11].

The expansion of the Universe can be observed through the cosmological redshift. Due to the expansion of space, the wavelength of photons emitted by distant objects is stretched during their propagation. The redshift is defined as

$$z = \frac{\lambda - \lambda_0}{\lambda_0},$$

where λ_0 is the emitted wavelength and λ is the observed wavelength. The redshift is directly related to the scale factor through

$$1 + z = \frac{a(t_0)}{a(t_e)} \tag{2.1}$$

with t_e denoting the time of emission. Consequently, the observed redshift provides information about the expansion history of the Universe and can be used to determine cosmological distances. The line-of-sight comoving distance to an object at redshift z is given by

$$\chi(z) = \chi_H \int_0^z \frac{dz'}{E(a(z'))},$$

where $\chi_H = c/H_0$ is the Hubble distance and $E(a) = H(a)/H_0$ is the dimensionless Hubble parameter [9].

2.2 Friedmann Equations and Cosmic Expansion

Following the cosmological principle, the matter and energy content of the Universe can be described by a perfect-fluid energy-momentum tensor. In combination with the FLRW metric, Einstein's field equations reduce to the *Friedmann equations*

$$H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda}{3}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda}{3}$$

which determine the evolution of the scale factor $a(t)$. Here ρ and p denote the energy density and pressure of the cosmic fluid, Λ is the cosmological constant, and G is the gravitational constant [10].

The critical density at present time is defined as the total energy density for which the Universe is spatially flat ($k = 0$) and is given by,

$$\rho_c = \frac{3H_0^2}{8\pi G}.$$

The ratio between the density of a cosmological component and the critical density defines the corresponding density parameter,

$$\Omega_i = \frac{\rho_i}{\rho_c}.$$

with Ω_m , Ω_r , Ω_Λ , and Ω_k denoting the density parameters of matter, radiation, vacuum (dark) energy, and curvature, respectively.

Consequently, the Friedmann equation can be expressed in terms of the density parameters [1]:

$$H^2 = H_0^2 [\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda].$$

Using Equation 2.1, the dimensionless Hubble parameter can also be written as

$$E(z) = \sqrt{\Omega_{r,0}(1+z)^4 + \Omega_{m,0}(1+z)^3 + \Omega_{k,0}(1+z)^2 + \Omega_{\Lambda,0}}.$$

2.2.1 The Λ CDM Model

The current standard model of cosmology is the so-called Λ CDM model. It describes the evolution of the Universe assuming the presence of cold dark matter (CDM) and a cosmological constant Λ , which is associated with dark energy and the accelerated expansion of the Universe. The model is in good agreement with a wide range of cosmological observations [21]. For the late-time Universe (low redshift) probed by extragalactic sources, the radiation density is negligible and the expansion history can be approximated by the contributions from matter, curvature, and dark energy. Assuming a spatially flat Universe, the curvature term also vanishes, yielding [7]

$$E(z) \approx \sqrt{\Omega_{m,0}(1+z)^3 + \Omega_{\Lambda,0}}.$$

2.3 Fast Radio Bursts

Fast Radio Bursts (FRB) are bright, transient radio pulses with durations of milliseconds or less and peak flux densities ranging from 50 mJy to 100 Jy [23]. They have been detected across a frequency range of 110 MHz [25] to 8 GHz [38].

The isotropic-equivalent energy distribution of the observed FRB population follows a power-law distribution described by

$$N(E_{\text{iso}}) \propto E_{\text{iso}}^{-1.8},$$

indicating that high-energy bursts are increasingly rare compared to lower-energy events. The negative power-law index of approximately -1.8 describes the decrease in the occurrence rate with increasing isotropic-equivalent energy.

Furthermore, the implied isotropic energies of FRBs are within a few orders of magnitude of those released in gamma-ray bursts and supernova explosions, suggesting that FRBs may originate from highly energetic astrophysical processes [23, 37].

Consequently, a broad range of progenitor models has been proposed, including neutron stars, magnetars, white dwarfs, black holes ultra-luminous X-ray sources, active galactic nuclei, and more exotic scenarios such as superconducting cosmic strings [24].

Despite releasing within a single millisecond an amount of energy comparable to what the Sun radiates over three days [23], FRBs are typically located at cosmological distances. As their radio emission propagates across the expanding Universe, it

spreads over an immense area, so that the signal received on Earth is roughly a thousand times weaker than that of a mobile phone located on Earth's Moon [32] [ZU POPULAERWISSENSCHAFTLICH ???].

The signal from FRBs is dispersed as it propagates through the ionized interstellar and intergalactic medium, causing lower-frequency components to arrive later than higher-frequency components. This makes FRBs appear as a frequency-dependent sweep in time, which can be described by the following relation:

$$DM = \frac{2\pi m_e c}{e^2} \Delta t \left(\frac{1}{\nu_{\text{low}}^2} - \frac{1}{\nu_{\text{high}}^2} \right)^{-1},$$

where ν_{low} and ν_{high} are the lower and higher observing frequencies, Δt is the time delay between the arrival of the two frequency components, c is the speed of light, e is the elementary charge, and m_e is the electron mass. The dispersion measure DM quantifies the integrated column density of free electrons between the source and the observer:

$$DM \propto \int_0^d dl n_e(l),$$

where n_e represents the free electron density and d the distance between the source and the observer [23].

The origin of FRBs is established through their large DM , which exceed the expected contribution from the Galactic interstellar medium along the corresponding line of sight. For a source located within the Milky Way, typical values predicted by Galactic electron-density models are of the order of $DM_{\text{MW}} \lesssim 50 \text{ pc cm}^{-3}$.

In contrast, observed FRBs often exhibit significantly larger values $DM_{\text{FRB}} \gtrsim 1000 \text{ pc cm}^{-3}$ [27].

The first FRB, later named the Lorimer Burst, was identified in archival pulsar survey data acquired in 2001 and published in 2007 [17].

Initially, only a small number of FRBs had been detected, leading to the assumption that they represented rare, singular events. The situation changed decisively with the commissioning of CHIME in 2018, whose wide field of view drove an explosion in detections and increased the known FRB population by several orders of magnitude. The latest CHIME/FRB catalog contains 4539 detected burst events originating from 3641 unique FRB sources, including 981 bursts from 83 known repeating sources [5]. As shown in Figure 2.1, the known FRB sources exhibit an approximately

isotropic distribution across the sky, consistent with an extragalactic population and suggesting no preferred spatial direction. The shown distribution is based on the FRB sample available up to 2022 and serves as an illustration of the large-scale spatial distribution of FRBs.

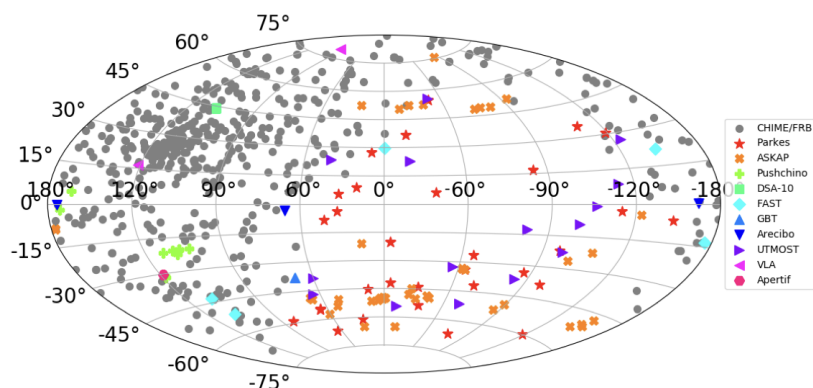


Figure 2.1: Aitoff projection of the known FRB population in Galactic coordinates based on observations available up to 2022. The sample is predominantly composed of CHIME/FRB detections, whose non-uniform sensitivity and sky coverage influence the observed distribution. After correcting for these observational biases, the underlying FRB population is consistent with an isotropic distribution expected for a cosmological source population [24].

2.3.1 Repeating FRBs

While the majority of FRBs have been observed as one-off events, a subset of sources has been found to emit multiple bursts over time. The existence of so-called repeaters immediately places a fundamental constraint on emission physics: any cataclysmic, single-event progenitor mechanism is directly ruled out for these sources, since the central engine must be capable of sustaining burst activity over timescales of years [23].

Some repeating FRBs also exhibit periodic modulation of their activity windows, as observed in FRB 20180916B [20], suggesting either a binary system geometry or precession of the emitting object. Whether all FRBs are intrinsically capable of repeating remains an open question [24].

2.3.2 Progenitor Mechanisms

A broad classification of proposed FRB progenitor models distinguishes between two main categories: catastrophic, one-off events and non-catastrophic scenarios involving long-lived central engines. Catastrophic models can account only for non-repeating FRBs and are therefore incompatible with the observed behavior of repeating sources. In contrast, non-catastrophic models allow for multiple burst episodes over extended periods. Among the proposed progenitors, compact stellar remnants are considered the most promising candidates. In particular, neutron stars and magnetars are strongly supported by both theoretical models and an increasing body of observational evidence [23].

Magnetars are neutron stars distinguished by surface magnetic fields many orders of magnitude stronger than those of ordinary radio pulsars, making them the most strongly magnetised stellar remnants known [23]. They arise primarily through the core collapse of massive stars in supernova explosions and are therefore intrinsically young objects whose formation rate is expected to closely mirror the cosmic star formation history (SFH), preferentially found in or near regions of active star formation [22]. The landmark detection in April 2020 of an FRB-like burst from the Galactic magnetar SGR 1935+2154 provided the most direct empirical link between magnetars and FRBs [4]. Population synthesis work using the CHIME/FRB catalogue finds that the hypothesis of FRBs tracking the SFH is not ruled out by current data, with a small delay preferred, consistent with a scenario in which young magnetars formed through core-collapse supernovae serve as the dominant progenitor class [19].

Neutron stars formed through delayed channels represent a second important progenitor class, motivated by FRBs localised to environments inconsistent with active star formation. FRBs 180924 and 190523, found in massive, low-star-formation-rate galaxies with large offsets from the host centre, display environments similar to those of short gamma-ray burst hosts associated with mergers of binary neutron-star or neutron star-black hole binaries [33]. The globular cluster localisation of FRB 20200120E reinforces this point, as such an old environment is generally not associated with recent star formation. Isolated core-collapse supernovae occur within approximately 3 Myr to 48 Myr after a burst of star formation, whereas binary interactions can delay some core-collapse events to about 200 Myr [36]. Accretion-induced collapse (AIC) and compact-object merger channels exhibit much broader delay-time distributions extending to several gigayears [29]. As a consequence, this progenitor population does not closely track the cosmic SFH but evolves more slowly with redshift [16].

These two progenitor classes predict observably distinct redshift distributions [16],

forming the theoretical basis for the host-population bias modelling adopted in this work, in which each class is assigned a redshift-dependent bias $b(z)$ encoding how strongly its occurrence rate evolves relative to the cosmic star-formation rate. The parameterisation and specific values are detailed in chapter 3.

WAS IST MIT DEN FRAGEN AN KLARA WEGEN DER EDGE CASES SIEHE SLACK?????

2.4 Galaxy Surveys and Catalogs

2.4.1 The Kilo-Degree Survey

The Kilo-Degree Survey (KiDS) is a wide-field optical imaging observation conducted with the 2.6 m VLT Survey Telescope (VST) at ESO’s Paranal Observatory, covering 1347 deg^2 complemented by the VIKING survey to create a nine-band optical-to-near-infrared survey [12, 35]. The data used in this work are drawn from the fifth and final data release (DR5, KiDS-Legacy), which achieves a 5σ limiting magnitude of 24.8 mag in the r -band with a median seeing of 0.7 arcsec [35, 13]. Photometric redshifts are calibrated using approximately 126000 spectroscopically confirmed sources [34].

2.5 Structure Formation

Within the Λ CDM framework, the large-scale structure of the Universe emerged from the gravitational amplification of small primordial density perturbations. According to the gravitational instability paradigm, quantum fluctuations seeded during an inflationary phase were stretched to cosmological scales and subsequently grew into the filaments, voids, and galaxy clusters observed today [2, 28]. The growth of perturbations depends sensitively on the cosmic expansion history, and the overall amplitude of matter fluctuations at the present time is conventionally parameterised by σ_8 [1].

2.6 Correlation Functions and Angular Power Spectrum

The statistical properties of the large-scale matter distribution are quantified through correlation functions and their Fourier-space counterparts, the power spectra.

The density contrast $\delta(\mathbf{x})$ is defined as the fractional deviation of the local matter density $\rho(\mathbf{x})$ from the mean cosmic density $\bar{\rho}$:

$$\delta(\mathbf{x}) := \frac{\rho(\mathbf{x}) - \bar{\rho}}{\bar{\rho}}.$$

The two-point correlation function is defined as

$$\xi(\mathbf{y}) := \langle \delta(\mathbf{x}) \delta(\mathbf{x} + \mathbf{y}) \rangle,$$

where the angle brackets denote an ensemble average. Statistical homogeneity implies that ξ depends only on the separation $\mathbf{y}$ and not on the absolute position $\mathbf{x}$, while statistical isotropy further implies that it depends only on the magnitude $y = |\mathbf{y}|$.

For a discrete tracer population, the two-point correlation function can be interpreted as describing the excess probability of finding a pair of tracers at a given separation relative to an unclustered random distribution. To describe the Fourier-transform pair of $\delta(\mathbf{x})$ and $\hat{\delta}(\mathbf{k})$, we adopt the convention

$$\hat{\delta}(\mathbf{k}) = \int d^3x \delta(\mathbf{x}) e^{-i\mathbf{k}\cdot\mathbf{x}}; \quad \delta(\mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \hat{\delta}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{x}}.$$

The three-dimensional matter power spectrum is then defined through

$$\langle \hat{\delta}(\mathbf{k}) \hat{\delta}^*(\mathbf{k}') \rangle = (2\pi)^3 \delta_{\text{D}}(\mathbf{k} - \mathbf{k}') P(k),$$

where δ_{D} denotes the Dirac delta function, reflecting statistical homogeneity in Fourier space. The power spectrum depends only on the magnitude $k = |\mathbf{k}|$, implying that it is isotropic in Fourier space [1].

When the tracers of interest are observed as projected distributions on the celestial sphere, it is common to describe the fields using a spherical-harmonic decomposition. On sufficiently large angular scales, a flat-sky Fourier transform is not generally adequate because of the curvature of the celestial sphere. A scalar field on the sphere can instead be expanded as

$$f(\hat{\mathbf{n}}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{\mathbf{n}}),$$

where $\hat{\mathbf{n}}$ denotes the direction on the sky and $Y_{\ell m}$ are the spherical harmonics. The corresponding harmonic coefficients are given by

$$a_{\ell m} = \int d\Omega f(\hat{\mathbf{n}}) Y_{\ell m}^*(\hat{\mathbf{n}}).$$

The characteristic angular scale associated with a multipole ℓ is approximately ℓ^{-1} [26]. For a statistically isotropic field, the angular power spectrum is defined through the covariance of the harmonic coefficients,

$$\langle a_{\ell m} a_{\ell' m'}^* \rangle = C_\ell \delta_{\ell\ell'} \delta_{mm'}.$$

An unbiased estimator of the angular power spectrum for an ideal full-sky observation is

$$\hat{C}_\ell = \frac{1}{2\ell + 1} \sum_{m=-\ell}^{\ell} |a_{\ell m}|^2.$$

For a finite or incomplete sky, additional effects such as the survey mask and mode coupling have to be taken into account [8].

The cross-correlation between two distinct projected fields A and B is described by the angular cross-power spectrum,

$$\langle a_{\ell m}^A a_{\ell' m'}^{B*} \rangle = C_\ell^{AB} \delta_{\ell\ell'} \delta_{mm'}.$$

A non-zero cross-power spectrum indicates statistical correlation between the two fields on the corresponding angular scales. For projected tracer fields, the angular fields can be written as line-of-sight integrals over the corresponding three-dimensional density contrasts,

$$\begin{aligned} A(\hat{\mathbf{n}}) &= \int d\chi W^A(\chi) \delta^A(\chi \hat{\mathbf{n}}), \\ B(\hat{\mathbf{n}}) &= \int d\chi W^B(\chi) \delta^B(\chi \hat{\mathbf{n}}), \end{aligned}$$

where χ denotes the comoving radial distance and $W^A(\chi)$ and $W^B(\chi)$ are the corresponding radial kernel functions. Using the spherical-harmonic expansion together with the Fourier representation of the three-dimensional fields, the full-sky angular cross-power spectrum can be written as

$$C_\ell^{AB} = \frac{2}{\pi} \int_0^\infty k^2 dk \int d\chi W^A(\chi) j_\ell(k\chi) \int d\chi' W^B(\chi') j_\ell(k\chi') P_{AB}(k; \chi, \chi'),$$

where j_ℓ denotes the spherical Bessel function and P_{AB} is the three-dimensional cross-power spectrum of the two tracer fields. The Limber approximation states that for approximately $\ell \gtrsim \mathcal{O}(10)$, the double integral collapses to a single line-of-sight integral [15]:

$$C_\ell^{AB} \approx \int \frac{d\chi}{\chi^2} W^A(\chi) W^B(\chi) P_{AB} \left(k = \frac{\ell + 1/2}{\chi}, z(\chi) \right).$$

The use of $\ell + 1/2$, rather than simply ℓ , provides a more accurate version of the Limber approximation, particularly at moderately large multipoles [18].

The angular cross-power spectrum therefore depends on the radial overlap of the two tracer populations, their relation to the underlying matter density field through their bias factors, and the shape and redshift evolution of the underlying matter power spectrum.

Consequently, the FRB-galaxy angular cross-power spectrum provides a sensitive probe of the spatial relation between FRBs and galaxies and can be used to constrain the redshift distribution, clustering properties, and host-population characteristics of FRBs [26, 31].

3 Methodology

Irgendwo das hier: Since luminous tracers such as galaxies do not perfectly follow the underlying dark matter distribution, their spatial statistics are related to the matter field through a bias parameter b , a concept that is formalised in section 2.6 and forms a central ingredient of this work [6].

3.1 Redshift Distribution

3.1.1 Shallow vs. Deep

3.2 Background Correlation

3.3 Auto-Correlation Analysis

3.4 Cross-Correlation Analysis

4 Results

4.1 Fisher Matrix

5 Conclusion & Outlook

6 Structure of the thesis

A possible structure for the thesis is as follows:

1. **Introduction**

The *short* introduction should present the motivation for the work and provide an overview of the upcoming chapters.

2. **Theoretical Background**

Keep theoretical background concise; refer to textbooks for standard material.

3. **Results**

The main part of the thesis where the conducted work and findings are presented.

4. **Summary and Outlook**

Summary of findings, possible improvements, and future directions.

Use chapters for the top-level structure. Avoid chapters with only one section.

Diese Vorlage ist auf die Kompilierung mit `lualatex` ausgelegt. Als Dokumentenklasse wird die KOMA-ScriptKlasse `scrbook` verwendet. Falls Sie Änderungen am Layout vornehmen möchten, lesen Sie die KOMA-Script-Dokumentation:

Eine umfangreiche Einführung in die moderne Verwendung von $\text{\LaTeX}$ gibt es hier: , lesenswert ist außerdem das $\text{\LaTeX}$ -Tabu

Um dieses Dokument vollständig zu erstellen sind maximal vier Programmläufe nötig:

1. `lualatex BachelorArbeit.tex`
2. `biber BachelorArbeit.bcf`
3. `lualatex BachelorArbeit.tex`
4. `lualatex BachelorArbeit.tex`

Beim ersten Lauf des $\text{\LaTeX}$ -Compilers werden die Kapitel, Links und zitierten Bibliographieeinträge in Hilfsdateien geschrieben.

Dann ist ein Lauf des Programms `biber` nötig, welches die benötigten Einträge aus der Hilfsdatei einliest, die Einträge aus der `.bib` Datei einliest, sortiert und formatiert und in eine weitere Hilfsdatei schreibt.

Beim nächsten L^AT_EX-Lauf werden dann diese Hilfsdateien eingelesen und Literatur- und Inhaltsverzeichnis erstellt.

Manchmal ist ein vierter Lauf nötig, falls sich durch das Einfügen des Literaturverzeichnisses Seitenzahlen verändert haben.

Das Tool `latexmk` übernimmt dies mit nur einem Programmaufruf und führt nur so viele Aufrufe durch, wie nötig sind.

```
latexmk --lualatex BachelorArbeit.tex
```

Eine gute Option ist es, den L^AT_EX Output in einem anderen Ordner zu erzeugen, dies ist mit der `--output-directory` Option möglich:

```
latexmk --output-directory=build --lualatex BachelorArbeit.tex
```

6.1 Erstellen des Ausgabedokuments mit Make

Für diese Vorlage wird ein Makefile zur Verfügung gestellt, welches automatisch alle Schritte ausführt, die für das fertige Dokument nötig sind. Die Ausgabe erfolgt dabei in den Unterordner `build/`. Make prüft, ob die Quelldateien verändert wurden, falls nicht, werden auch keine Befehle ausgeführt.

Falls Sie das Makefile benutzen möchten, sollten Sie alle Abhängigkeiten eintragen (Eigene Dateien für Kapitel, Plots, etc.).

Download und weitere Informationen zu Make gibt es unter . Die Befehle sind für die Bash ausgelegt. Wenn Sie sie unter Windows nutzen wollen, benötigen Sie einen Bash-Emulator, wie Git Bash, Download unter möglich. Wenn Sie Make installiert haben, rufen Sie einfach in der Konsole im Verzeichnis der Arbeit den Befehl `make`.

6.2 Erstellen des Ausgabedokuments mit Texmaker

6.2.1 Einrichten der nötigen Befehle

Ein beliebter Editor für alle Betriebssysteme ist Texmaker, Download unter . Damit Texmaker das Dokument korrekt kompiliert, fügen Sie einen benutzerdefinierten Befehl hinzu:

1. Klicken Sie oben in der Menüleiste auf *Benutzer/in*
2. Klick auf *Eigene Befehle*

3. Klick auf *Eigene Befehle editieren*, dort können Sie bis zu 5 eigene Befehle definieren
4. Geben Sie dem Befehl unter *Menüeintrag* einen Namen und tragen sie folgende Befehle in das Befehlsfeld ein:
`latexmk --lualatex --interaction=batchmode --halt-on-error %.tex |`
5. Bestätigen Sie mit *OK*

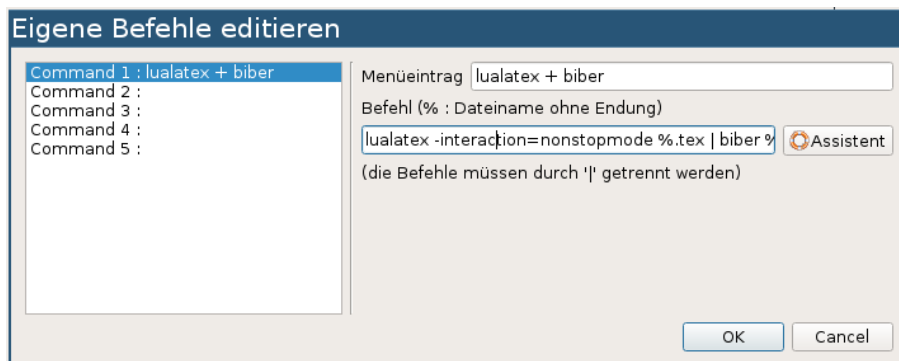


Figure 6.1: Screenshot zur Erstellung des Kompilier-Befehls in Texmaker

In Abbildung 6.1 ist ein Screenshot des Befehlsmenü gezeigt. Ihren Befehl können Sie nun im Drop-Down-Menü zum Kompilieren des Dokuments auswählen und mit einem Klick auf den Pfeil starten.

6.2.2 Aufräumen

Nach einem L^AT_EX-Fehler ist es oft notwendig, die erstellten Hilfsdateien zu löschen. Klicken Sie hierzu auf *Werkzeuge*→*Aufräumen*.

Bitte beachten Sie beim Schreiben der Arbeit folgende Konventionen bzw. Grundlagen:

- **Abschnitte und Zeilenumbrüche**

Es sollten im Fließtext keine Zeilenumbrüche mit `\\` erzwungen werden. Schreiben Sie höchstens einen Satz in eine Code-Zeile. Absätze werden im Code mit einer Leerzeile markiert und dann entsprechend der Einstellung von `parskip` in der Dokumentenklasse gesetzt.

- **Kursiv/Aufrecht**

- Variablen und physikalische Größen werden kursiv gesetzt.

- Einheiten werden immer aufrecht und mit einem halben Leerzeichen Abstand zur Zahl gesetzt. Nutzen Sie `siunitx`!
- Mathematische Konstanten und Funktionen werden ebenfalls aufrecht gesetzt. Zum Beispiel die Eulersche Zahl e , das imaginäre i und das infinitesimale d . Im Mathematikmodus können Sie dies mit dem Befehl `\mathrm{}` erreichen. Für die Funktionen stellt $\LaTeX$ Befehle bereit, z.B. `\arccos`.
- Integrand und ein dx sollten ebenfalls durch ein kleines Leerzeichen (`\,`) getrennt werden.

6.3 Zahlen und Einheiten

Jede Zahl, jede Einheit und jede Zahl mit Einheit sollte mit Hilfe der in dem Paket `siunitx` zur Verfügung gestellten Befehle gesetzt werden. Grundsätzlich gilt: Einheiten werden aufrecht gesetzt und haben ein kleines Leerzeichen (`\,`) Abstand zu ihrer Zahl. Werden Fließkommazahlen ohne `siunitx` gesetzt, entsteht ein hässlicher Leerraum zwischen Komma und erster Nachkommastelle, da $\LaTeX$ das Komma nicht als Dezimaltrennzeichen, sondern als Satzzeichen interpretiert.

Das Paket wurde mit deutschen Spracheinstellungen (also mit Komma als Dezimaltrennzeichen und $\cdot$ zwischen Zahl und Zehnerpotenz) geladen, sowie mit den Einstellungen, dass die Standardabweichung stets durch $\pm$ abgetrennt wird und Einheiten falls nötig als Brüche ausgegeben werden.

Table 6.1: Beispiele für `siunitx`

Befehl	Ergebnis
<code>\num{1.2345}</code>	1.2345
<code>\num{1.2e3}</code>	1.2×10^3
<code>\num{1.2 +- 0.2}</code>	1.2 ± 0.2
<code>\num{10000}</code>	10 000
<code>\si{\meter\per\second}</code>	m/s
<code>\SI{1.2(1)}{\micro\ampere}</code>	$(1.2 \pm 0.1) \mu\text{A}$
<code>\SI{1.2\pm0.1e3}{\kilo\gram\per\cubic\meter}</code>	$(1.2 \pm 0.1) \times 10^3 \text{ kg/m}^3$

Das Paket stellt unter anderem die drei wichtigen Befehle

- `\num{Zahl}`,
- `\si{Einheit}` und

- `\SI{Zahl}{Einheit}`

zur Verfügung. Diese Befehle sollten stets genutzt werden, wenn Zahlen angegeben werden. Sie funktionieren sowohl im Text- als auch im Mathematikmodus. In Tabelle 6.1 sind einige Beispiele aufgetragen. Bitte lesen Sie die Dokumentation .

6.4 Das Literaturverzeichnis

Im Text können Sie mit `\cite{kürzel}` zitieren. Seitenzahlen geben Sie in eckigen Klammern an: `\cite[10]{kürzel}`.

6.5 Figures

When preparing plots, use sufficiently large axis labels, bold enough font weights and distinguishable colors. Ideally, plots should use the same font and font size as the document. Create plots at the correct size and include them with the optional argument `scale=1` when possible.

Prefer vector graphics (PDF) and use raster images (PNG, JPG) only when necessary. End figure captions with a period.

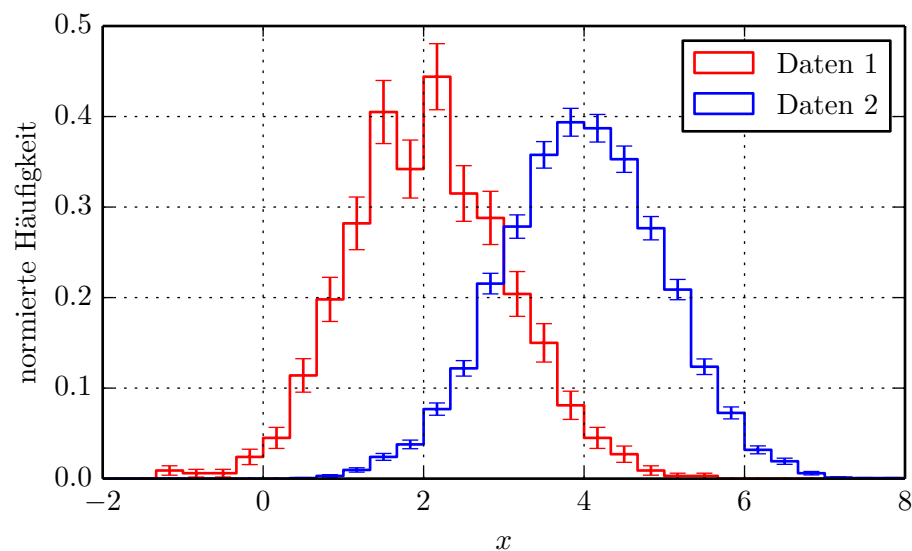


Figure 6.2: An example histogram with error bars for two datasets; font size and style match the document.

6.6 Tables

Keep tables as simple as possible and avoid excessive rules. In most cases three horizontal rules are sufficient: above, below and between the header and the body. The `booktabs` package provides `\toprule`, `\midrule` and `\bottomrule` for this purpose. The `siunitx` package supplies the `S` column type for nicely aligned numbers.

An example is Table 6.2. ngerman

Table 6.2: Example table using the `S` column type from `siunitx`.

p / Pa	T / K
1024.23	273.15
1025.31	274.5
1026.27	276.2

A Appendix

This appendix can contain supplementary material such as code listings, drawings or other extended material that does not fit into the main text. In most cases, all relevant results should be presented in the main part of the thesis and an appendix is optional.

Bibliography

- [1] Matthias Bartelmann and Peter Schneider. “Weak Gravitational Lensing.” In: *Physics Reports* 340 (2001), pp. 291–472. DOI: 10.1016/S0370-1573(00)00082-X. arXiv: astro-ph/9912508 [astro-ph].
- [2] Francis Bernardeau et al. “Large-scale structure of the Universe and cosmological perturbation theory.” In: *Physics Reports* 367 (2002), pp. 1–248. DOI: 10.1016/S0370-1573(02)00135-7. arXiv: astro-ph/0112551 [astro-ph].
- [3] Jorge L. Cervantes-Cota et al. “Cosmology today—A brief review.” In: *AIP Conference Proceedings*. AIP, 2011, pp. 28–52. DOI: 10.1063/1.3647524. URL: <http://dx.doi.org/10.1063/1.3647524>.
- [4] CHIME/FRB Collaboration. “A bright millisecond-duration radio burst from a Galactic magnetar.” In: *Nature* 587.7832 (2020), pp. 54–58. DOI: 10.1038/s41586-020-2863-y. arXiv: 2005.10324 [astro-ph.HE].
- [5] CHIME/FRB Collaboration. “The Second CHIME/FRB Catalog of Fast Radio Bursts.” In: *The Astrophysical Journal Supplement Series* 283.1 (2026), p. 34. DOI: 10.3847/1538-4365/ae3828. arXiv: 2601.09399 [astro-ph.HE].
- [6] Vincent Desjacques, Donghui Jeong, and Fabian Schmidt. “Large-scale galaxy bias.” In: *Physics Reports* 733 (2018), pp. 1–193. DOI: 10.1016/j.physrep.2017.12.002. arXiv: 1611.09787 [astro-ph.CO].
- [7] Yungui Gong, Xiao-ming Zhu, and Zong-Hong Zhu. “Current cosmological constraints on the curvature, dark energy and modified gravity.” In: *Monthly Notices of the Royal Astronomical Society* 415 (2011), pp. 1943–1950. DOI: 10.1111/j.1365-2966.2011.18846.x. arXiv: 1008.5010 [astro-ph.CO].
- [8] Eric Hivon et al. “MASTER of the Cosmic Microwave Background Anisotropy Power Spectrum.” In: *The Astrophysical Journal* 567 (2002), pp. 2–17. DOI: 10.1086/338126. arXiv: astro-ph/0105302 [astro-ph].
- [9] David W. Hogg. “Distance measures in cosmology.” In: (2000). arXiv: astro-ph/9905116 [astro-ph]. URL: <https://arxiv.org/abs/astro-ph/9905116>.

- [10] Dragan Huterer and Daniel L. Shafer. “Dark energy two decades after: Observables, probes, consistency tests.” In: *Reports on Progress in Physics* 81.1 (2018), p. 016901. DOI: 10.1088/1361-6633/aa997e. arXiv: 1709.01091 [astro-ph.CO].
- [11] Mustapha Ishak. “Testing General Relativity in Cosmology.” In: *Living Reviews in Relativity* 22.1 (2019), p. 1. ISSN: 1433-8351. DOI: 10.1007/s41114-018-0017-4. URL: <https://doi.org/10.1007/s41114-018-0017-4>.
- [12] Jelte T. A. de Jong et al. “The Kilo-Degree Survey.” In: *Experimental Astronomy* 35 (2013), pp. 25–44. DOI: 10.1007/s10686-012-9306-1. arXiv: 1206.1254 [astro-ph.CO].
- [13] KiDS Collaboration. *KiDS-ESO Data Release 5*. Kilo-Degree Survey, 2025. URL: <https://kids.strw.leidenuniv.nl/DR5/> (visited on 08/27/2026).
- [14] Pierre Laurent et al. “A $14 h^{-3} \text{ Gpc}^3$ Study of Cosmic Homogeneity Using BOSS DR12 Quasar Sample.” In: *Journal of Cosmology and Astroparticle Physics* 2016.11 (2016), p. 060. DOI: 10.1088/1475-7516/2016/11/060. arXiv: 1602.09010 [astro-ph.CO].
- [15] D. Nelson Limber. “The Analysis of Counts of the Extragalactic Nebulae in Terms of a Fluctuating Density Field.” In: *The Astrophysical Journal* 117 (1953), pp. 134–141. DOI: 10.1086/145661.
- [16] Hai-Nan Lin, Xin-Yi Li, and Rui Zou. “Time Delay of Fast Radio Burst Population with Respect to the Star Formation History.” In: *Monthly Notices of the Royal Astronomical Society* 514.2 (2022), pp. 2228–2236. DOI: 10.1093/mnras/stac1483. arXiv: 2205.11145 [astro-ph.HE].
- [17] Duncan R. Lorimer et al. “A Bright Millisecond Radio Burst of Extragalactic Origin.” In: *Science* 318.5851 (2007), pp. 777–780. DOI: 10.1126/science.1147532. arXiv: 0709.4301 [astro-ph].
- [18] Marilena LoVerde and Niayesh Afshordi. “Extended Limber approximation.” In: *Physical Review D* 78 (2008), p. 123506. DOI: 10.1103/PhysRevD.78.123506. arXiv: 0809.5112 [astro-ph].
- [19] Min Meng and Can-Min Deng. “Constraints on the progenitor models of fast radio bursts from population synthesis with the first CHIME/FRB catalog.” In: *Astronomy & Astrophysics* 698 (2025), A127. DOI: 10.1051/0004-6361/202451250. arXiv: 2506.04986 [astro-ph.HE].
- [20] Cherry Ng. “A Brief Review on Fast Radio Bursts.” In: *Proceedings of the SF2A 2023*. 2023, pp. 37–40. DOI: 10.48550/arXiv.2311.01899. arXiv: 2311.01899 [astro-ph.HE].

-
- [21] P. J. E. Peebles and Bharat Ratra. “The Cosmological Constant and Dark Energy.” In: *Reviews of Modern Physics* 75 (2003), pp. 559–606. DOI: 10.1103/RevModPhys.75.559. arXiv: astro-ph/0207347 [astro-ph].
- [22] Davide Pellicciari et al. “The Northern Cross Fast Radio Burst project: III. The FRB–magnetar connection in a sample of nearby galaxies.” In: *Astronomy & Astrophysics* 674 (2023), A223. DOI: 10.1051/0004-6361/202346307. arXiv: 2304.11179 [astro-ph.HE].
- [23] Emily Petroff, J. W. T. Hessels, and Duncan R. Lorimer. “Fast Radio Bursts.” In: *The Astronomy and Astrophysics Review* 27.1 (2019), p. 4. DOI: 10.1007/s00159-019-0116-6. arXiv: 1904.07947 [astro-ph.HE].
- [24] Emily Petroff, J. W. T. Hessels, and Duncan R. Lorimer. “Fast radio bursts at the dawn of the 2020s.” In: *The Astronomy and Astrophysics Review* 30.1 (2022), p. 2. DOI: 10.1007/s00159-022-00139-w. arXiv: 2107.10113 [astro-ph.HE].
- [25] Ziggy Pleunis et al. “LOFAR Detection of 110–188 MHz Emission and Frequency-dependent Activity from FRB 20180916B.” In: *The Astrophysical Journal Letters* 911.1 (2021), p. L3. DOI: 10.3847/2041-8213/abec72. arXiv: 2012.08372 [astro-ph.HE].
- [26] Masoud Rafiei-Ravandi et al. “CHIME/FRB Catalog 1 Results: Statistical Cross-correlations with Large-scale Structure.” In: *The Astrophysical Journal* 922 (2021), p. 42. DOI: 10.3847/1538-4357/ac1dab. arXiv: 2106.04354 [astro-ph.CO].
- [27] Robert Reischke, Steffen Hagstotz, and Robert Lilow. “Probing primordial non-Gaussianity with fast radio bursts.” In: *Physical Review D* 103.2 (2021), p. 023517. DOI: 10.1103/PhysRevD.103.023517. arXiv: 2007.04054 [astro-ph.CO].
- [28] Antonio Riotto. *Inflation and the Theory of Cosmological Perturbations*. 2003. arXiv: hep-ph/0210162 [hep-ph].
- [29] A. J. Ruiter et al. “On the formation of neutron stars via accretion-induced collapse in binaries.” In: *Monthly Notices of the Royal Astronomical Society* 484.1 (2019), pp. 698–711. DOI: 10.1093/mnras/stz001. arXiv: 1802.02437 [astro-ph.SR].
- [30] M. I. Scrimgeour et al. “The WiggleZ Dark Energy Survey: the transition to large-scale cosmic homogeneity.” In: *Monthly Notices of the Royal Astronomical Society* 425.1 (2012), pp. 116–134. DOI: 10.1111/j.1365-2966.2012.21402.x. arXiv: 1205.6812 [astro-ph.CO].

- [31] Masato Shirasaki, Kazumi Kashiya, and Naoki Yoshida. “Large-scale clustering as a probe of the origin and the host environment of fast radio bursts.” In: *Physical Review D* 95 (2017), p. 083012. DOI: 10.1103/PhysRevD.95.083012. arXiv: 1702.07085 [astro-ph.CO].
- [32] Space.com. *Fast Radio Burst Repeater Discovered in New Observation*. Accessed: 2026-07-31. 2018. URL: <https://www.space.com/42943-fast-radio-burst-repeater-new-discovery.html>.
- [33] Fa-Yin Wang et al. “Fast Radio Bursts from Activities of Neutron Stars Newborn in BNS Mergers: Offset, Birth Rate, and Observational Properties.” In: *The Astrophysical Journal* 891.1 (2020), p. 72. DOI: 10.3847/1538-4357/ab74d0. arXiv: 2002.03507 [astro-ph.HE].
- [34] Angus H. Wright et al. “KiDS-Legacy: Redshift distributions and their calibration.” In: *Astronomy & Astrophysics* 703 (2025), A144. DOI: 10.1051/0004-6361/202554909. arXiv: 2503.19440 [astro-ph.CO].
- [35] Angus H. Wright et al. “The fifth data release of the Kilo-Degree Survey: Multi-epoch optical/NIR imaging covering wide and legacy-calibration fields.” In: *Astronomy & Astrophysics* 686 (2024), A170. DOI: 10.1051/0004-6361/202346730. arXiv: 2503.19439 [astro-ph.GA].
- [36] E. Zapartas et al. “Delay-time distribution of core-collapse supernovae with late events resulting from binary interaction.” In: *Astronomy & Astrophysics* 601 (2017), A29. DOI: 10.1051/0004-6361/201629685. arXiv: 1702.03979 [astro-ph.SR].
- [37] Rachel C. Zhang et al. “On the energy and redshift distributions of fast radio bursts.” In: *Monthly Notices of the Royal Astronomical Society* 501.1 (2021), pp. 157–167. DOI: 10.1093/mnras/staa3537. arXiv: 2011.06151 [astro-ph.HE].
- [38] Yunfan Gerry Zhang et al. “Fast Radio Burst 121102 Pulse Detection and Periodicity: A Machine Learning Approach.” In: *The Astrophysical Journal* 866.2 (2018), p. 149. DOI: 10.3847/1538-4357/aadf31. arXiv: 1809.03043 [astro-ph.HE].

Hinweis zur Formulierungshilfe: Einzelne Formulierungen dieser Arbeit wurden mit Unterstützung von ChatGPT (OpenAI) überarbeitet. Wissenschaftliche Inhalte, Auswertungen und Schlussfolgerungen liegen vollständig bei der Autorin/dem Autor; für verbleibende Fehler trägt die Autorin/der Autor die Verantwortung.

Eidesstattliche Versicherung

(Affidavit)

Name, Vorname
(surname, first name)

Matrikelnummer
(student ID number)

☐ Bachelorarbeit
(Bachelor's thesis)

☐ Masterarbeit
(Master's thesis)

Titel
(Title)

Ich versichere hiermit an Eides statt, dass ich die vorliegende Abschlussarbeit mit dem oben genannten Titel selbstständig und ohne unzulässige fremde Hilfe erbracht habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate kenntlich gemacht. Die Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen.

I declare in lieu of oath that I have completed the present thesis with the above-mentioned title independently and without any unauthorized assistance. I have not used any other sources or aids than the ones listed and have documented quotations and paraphrases as such. The thesis in its current or similar version has not been submitted to an auditing institution before.

Ort, Datum
(place, date)

Unterschrift
(signature)

Belehrung:

Wer vorsätzlich gegen eine die Täuschung über Prüfungsleistungen betreffende Regelung einer Hochschulprüfungsordnung verstößt, handelt ordnungswidrig. Die Ordnungswidrigkeit kann mit einer Geldbuße von bis zu 50.000,00 € geahndet werden. Zuständige Verwaltungsbehörde für die Verfolgung und Ahndung von Ordnungswidrigkeiten ist der Kanzler/die Kanzlerin der Technischen Universität Dortmund. Im Falle eines mehrfachen oder sonstigen schwerwiegenden Täuschungsversuches kann der Prüfling zudem exmatrikuliert werden. (§ 63 Abs. 5 Hochschulgesetz - HG -).

Die Abgabe einer falschen Versicherung an Eides statt wird mit Freiheitsstrafe bis zu 3 Jahren oder mit Geldstrafe bestraft.

Die Technische Universität Dortmund wird ggf. elektronische Vergleichswerkzeuge (wie z.B. die Software „turnitin“) zur Überprüfung von Ordnungswidrigkeiten in Prüfungsverfahren nutzen.

Die oben stehende Belehrung habe ich zur Kenntnis genommen:

Official notification:

Any person who intentionally breaches any regulation of university examination regulations relating to deception in examination performance is acting improperly. This offense can be punished with a fine of up to EUR 50,000.00. The competent administrative authority for the pursuit and prosecution of offenses of this type is the Chancellor of TU Dortmund University. In the case of multiple or other serious attempts at deception, the examinee can also be unenrolled, Section 63 (5) North Rhine-Westphalia Higher Education Act (*Hochschulgesetz, HG*).

The submission of a false affidavit will be punished with a prison sentence of up to three years or a fine.

As may be necessary, TU Dortmund University will make use of electronic plagiarism-prevention tools (e.g. the "turnitin" service) in order to monitor violations during the examination procedures.

I have taken note of the above official notification:*

Ort, Datum
(place, date)

Unterschrift
(signature)

***Please be aware that solely the German version of the affidavit ("Eidesstattliche Versicherung") for the Bachelor's/ Master's thesis is the official and legally binding version.**